To prove:

$\forall m : nat. \quad 0 + m = m$

$$\boxed{\begin{array}{l} \\ n + m = \begin{cases} m & n = 0 \\ 1 + (n' + m) & n = 1 + n' \end{cases} \end{array}}$$

Let m be a natural number.

We need to show $0 + m = m$

From the definition of "+" we know that $0+m$ computes to m.

So it is left to show that $m = m$.

By the reflexivity of equality $m = m$ hold.

Qed. $\square$.

To prove:

$\forall\, n\ m : nat.\ \underbrace{n = m \to n + n = m + m}$

Let n and m be natural numbers.

We show that $\underbrace{\text{if } n = m}$ $\underbrace{\text{then } n + n = m + m}$.

Assume $n = m$ (*).

We show that $\overset{m}{n} + \overset{m}{n} = m + m$

Using (*) we can replace n with m

We are left to show $m + m = m + m$.

By reflexivity of "=" this holds $\blacksquare$